

Supplementary Materials for C²GES: Structure-Aware Extractive Summarization of Long Power-System Technical Reports with Typed-Path Graphs

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Contents and Data Files

This supplement separates scientific interpretation from implementation records. It contains four tables referenced by the main manuscript. Paths below are relative to the C2GES release root. Machine-readable files are provided under `03_Reproducibility/Data/`: the non-verbatim sampling frame, aggregate metrics, paired ROUGE-L differences, output-length records, exact sign-flip results, page locators, and the development-only calibration report. Source PDFs and verbatim extracted passages are excluded because third-party redistribution permission has not been established.

Table S1. Rights-Safe Sampling Frame

The complete 40-row sampling frame is supplied as `03_Reproducibility/Data/rights_safe_metadata/rights_safe_report_metadata.csv`. Its 23 fields include the report identifier, source locator, split, series identifier, inclusion decision, source-file hash, and redistribution status; it contains no report passages.

Quantity	Value
Inspected complete PDFs	40
Declared pages	3200
Retained / excluded reports	27 / 13
Development / test reports	12 / 15
Non-summary candidates	12,924

Table S2. Output-Length Records

The complete 14 condition–budget summary is supplied as `03_Reproducibility/Data/postrun_diagnostics/output_length_summary.csv`; per-report values are in `03_Reproducibility/Data/postrun_diagnostics/output_length_per_report.csv`.

K	Condition	Mean words	Mean chars.	> 100	Table	Max. words
5	Full C ² GES	287.7	1843.6	12	14	140
5	Strict variant	329.0	2099.9	15	16	270
5	Semantic-MMR	184.7	1242.1	5	9	138
5	TextRank	177.0	1125.1	1	7	124
10	Full C ² GES	568.9	3619.5	25	26	270
10	Strict variant	596.9	3822.5	28	26	270
10	Semantic-MMR	354.5	2361.2	7	16	138
10	TextRank	369.0	2329.7	4	18	130

Table S3. Corpus and Implementation Checks

Checked property	Recorded result
Reference/body page-interval overlap	0
Normalized exact candidate/reference matches	0
Normalized common substrings of at least 50 characters	0
Report-condition-budget records / expected records	210 / 210
Selected source references resolved to a valid page	1575 / 1575
Maximum path expansions / configured limit	19,638 / 2,000,000
Maximum qualifying paths / configured limit	16,182 / 250,000
Full/strict score comparisons changed	9774 / 19,008
Report-budget selection sequences changed	28 / 30
Aggregate and paired-contrast recomputation	Exact agreement

Aggregate values are available at `03_Reproducibility/Data/audits/aggregate_metrics.json`; non-verbatim paired differences and page locators are available at `03_Reproducibility/Data/figure_inputs/paired_rougel_differences_nonverbatim.csv` and `03_Reproducibility/Data/postrun_diagnostics/selected_page_locator.csv`. Detailed checks are retained with the corresponding code and archived run records. Passing these checks establishes consistency of the reported computation; it does not determine semantic summary quality.

Table S4. Exact Paired Sign-Flip Sensitivity

For each contrast, all $2^{15} = 32,768$ report-level sign assignments were enumerated. The procedure assumes report-level independence and sign exchangeability. Holm adjustment treats the six values as one family.

<i>K</i>	Baseline	Signs + / - / 0	Exact <i>p</i>	Holm-adjusted <i>p</i>
5	Strict variant	7/7/1	0.436768	0.436768
5	Semantic-MMR	14/1/0	0.000305	0.001526
5	TextRank	14/1/0	0.000122	0.000732
10	Strict variant	6/8/1	0.200684	0.401367
10	Semantic-MMR	11/4/0	0.006592	0.026367
10	TextRank	13/2/0	0.009644	0.028931

These post-result values are sensitivity analyses. The main manuscript emphasizes effect sizes, paired distributions, and report-composition intervals.

The complete machine-readable result is `03_Reproducibility/Data/postrun_sensitivity/exact_signflip_results.csv`; the JSON file in the same directory retains the report-level non-verbatim deltas used in the calculation.

Exploratory Development-Only Calibration

The complete chronology, configuration space, leave-one-report-out results, and machine-readable records are described in `03_Reproducibility/Data/POST_UNBLINDING_DEV_CALIBRATION_SUPPLEMENT.md`. A zero path-deletion weight was selected in all 12 development folds; the analysis was not applied to the retained 15-report test set.

A separate equal-budget exercise is documented in `03_Reproducibility/Data/dev_balanced_tuning/`. Semantic-MMR, TextRank, and normalized-path C²GES each received nine configurations and the same ordered development objective over both extraction budgets. The selected values were 0.9, 0.65, and 0.0, respectively. The decision record states `test_input_accessed=false`; these configurations are authorized only for a future preregistered external-series comparison and do not alter the retained-test results.